SUMMARY NOTES

Photosynthesis Biochemical Mechanisms

EduForge AI • Generated Teaching Material • September 15, 2026

Executive Summary

Photosynthesis is the fundamental biochemical process by which photoautotrophic organisms synthesize organic compounds.

Core Reaction Pathways

- Light-dependent reactions occur within the thylakoid membrane.
- The Calvin cycle (light-independent reactions) occurs within the stroma.
- ATP and NADPH synthesized in the light phase drive carbon fixation.

Phase	Reaction Location	Key Inputs	Primary Products
Light Reactions	Thylakoid Membrane	H2O, Sunlight, NADP+, ADP	O2, ATP, NADPH
Calvin Cycle	Chloroplast Stroma	CO2, ATP, NADPH	G3P Sugar, ADP, NADP+

Key Takeaways

- Without photosynthesis, primary biomass production would cease.
- The process is regulated by ambient CO2 concentration, light intensity, and temperature.